DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture Week 5: Assignment

No-Arbitrage · Replicating Portfolios · Risk-Neutral Pricing

Instructions

- This assignment covers Lectures 13–15 and the Week 5 Lab. It carries **100 marks** total.
- Show all working for numerical questions – correct answers with no working receive partial credit only.
- Questions marked **[Python]** require a short script or notebook cell; submit your code alongside your written answers.
- Submit a single PDF (typeset or clearly scanned handwriting) plus any `.py`/`.ipynb` files, via the course submission portal.

Part A – Concepts (15 marks)

A1. (4 marks) Define arbitrage precisely, in terms of net investment, probability of loss, and probability of profit.

A2. (4 marks) State the Law of One Price, and explain the logical (not economic) basis for why it must hold.

A3. (4 marks) Explain why no-arbitrage pricing requires fewer assumptions than CAPM, and why this makes it a more robust foundation for pricing derivatives.

A4. (3 marks) True or false, with justification: "If the real-world probability of the stock going up is $p = 0.9$ (very likely), the risk-neutral probability q must also be close to 1."

Part B – Replicating Portfolios (25 marks)

B1. (10 marks) $S_0 = 80$, $u = 1.25$, $d = 0.8$, $r = 6\%$. A put option has strike $K = 85$. Compute C_u, C_d, then Δ, B, then C_0, showing every step.

B2. (5 marks) Explain why a put option's Δ is always between -1 and 0, in contrast to a call option's $\Delta \in [0, 1]$. Give an intuitive reason referencing how a put's payoff moves relative to the stock price.

B3. (10 marks) $S_0 = 200$, $u = 1.15$, $d = 0.95$, $r = 4\%$. A "digital" (binary) option pays a fixed Rs. 50 if the stock goes up, and Rs. 0 if it goes down. Compute Δ, B, C_0 for this option. (Note: Δ need not lie in $[0, 1]$ for exotic payoffs like this one – comment briefly on why, given how sharply this payoff jumps between states.)

Part C – Risk-Neutral Probabilities (20 marks)

C1. (8 marks) For the put option in B1, compute the risk-neutral probability q and re-price the put using risk-neutral valuation. Confirm it matches your answer to B1.

C2. (8 marks) Starting from $\Delta = (C_u - C_d)/(S_u - S_d)$ and $B = (C_d - \Delta S_d)/(1+r)$, substitute both into $C_0 = \Delta S_0 + B$ and simplify algebraically to derive the risk-neutral formula $C_0 = \frac{1}{1+r}[qC_u + (1-q)C_d]$ with $q = \frac{(1+r)-d}{u-d}$. Show every algebraic step (this is the derivation only sketched in Lecture 15 – write it out in full here).

C3. (4 marks) If $u = 1.3$ and $d = 0.7$, find the range of risk-free rates r for which this one-period model remains arbitrage-free.

Part D – Arbitrage Detection (20 marks)

D1. (10 marks) $S_0 = 120$, $u = 1.15$, $d = 0.85$, $r = 6\%$, call option strike $K = 125$. Compute the theoretical no-arbitrage price C_0. If this call is quoted in the market at Rs. 5.00, construct the explicit, exact arbitrage strategy (specify the exact stock and bond positions) and compute its riskless profit today.

D2. (5 marks) A trader says: "Since I'm confident this stock will go up, I should value the call option using a probability of 0.9, not the risk-neutral q." Explain why this reasoning is incorrect for the purpose of *pricing* the option (even though the trader's belief might be perfectly relevant for deciding whether to make a *directional bet* on the stock itself).

D3. (5 marks) For a model with $S_0 = 100$, $u = 1.1$, $d = 0.95$, and $r = 15\%$ (so $1 + r = 1.15 > u$), describe the explicit "short the stock, invest at r" arbitrage strategy, and compute its guaranteed profit in both the up and down states.

Part E – Computational (15 marks)

E1. (15 marks) [Python] Using your Lab code (or a fresh, self-contained script), fix $S_0 = 100$, $u = 1.2$, $d = 0.9$, $r = 5\%$, and compute the no-arbitrage call price C_0 (via replication) for strikes K ranging from 70 to 130 in steps of 2. Plot C_0 against K. Comment in 2–3 sentences on the shape of this curve: is it increasing or decreasing in K? Is it convex? Does this match your intuition for how a call option's value should depend on its strike?

Part F – Integrative Question (5 marks)

F1. (5 marks) Both β_i (Week 4, CAPM) and Δ (this week, replication) measure a kind of sensitivity of one asset's value to another's movements. In a short paragraph (80–120 words), explain what each one measures, and contrast *how* each is derived – one from statistical/equilibrium reasoning across many assets and much historical data, the other from a purely logical no-arbitrage argument within a single, fully specified one-period model.

End of Assignment. Solutions will be released after the submission deadline.